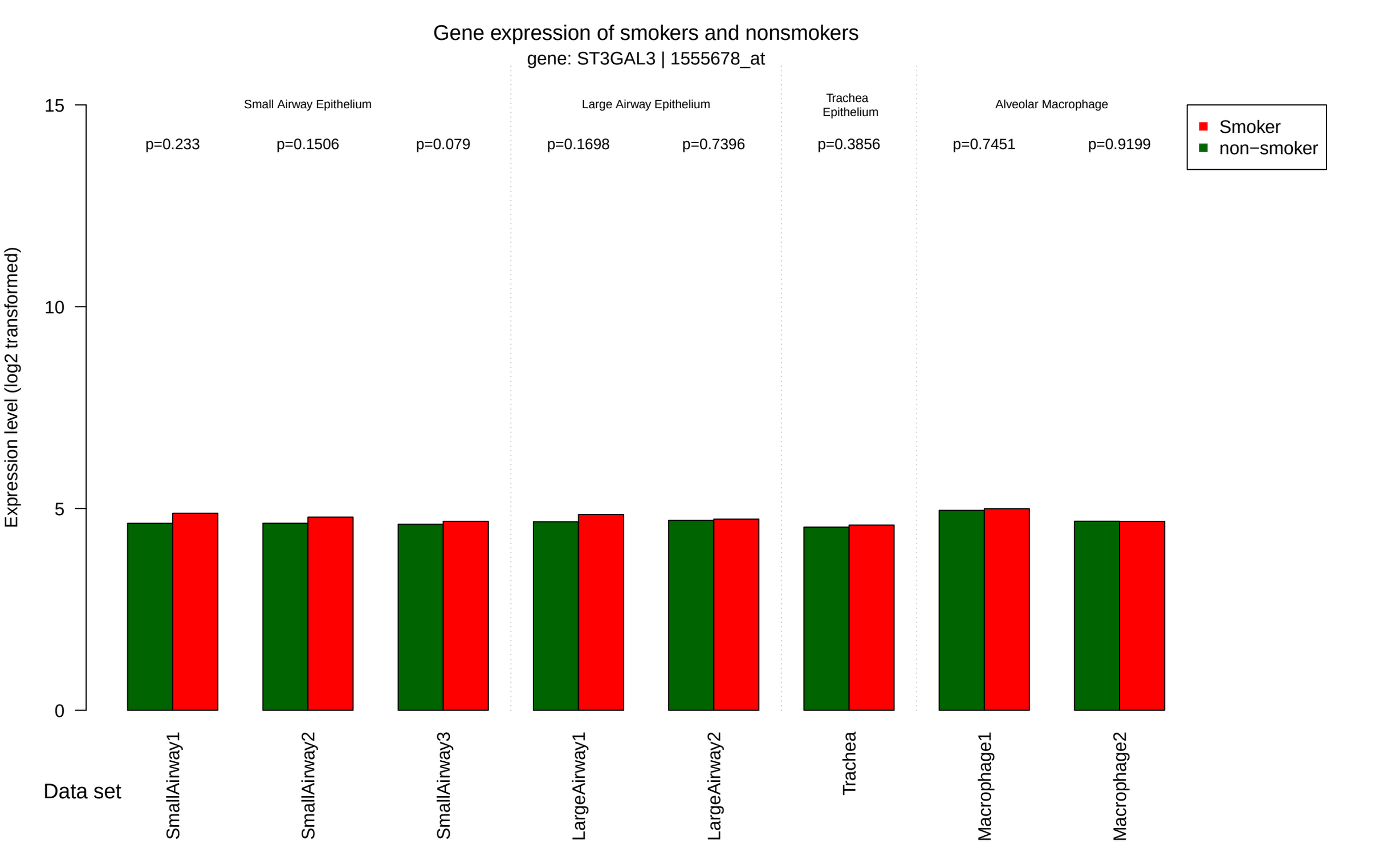
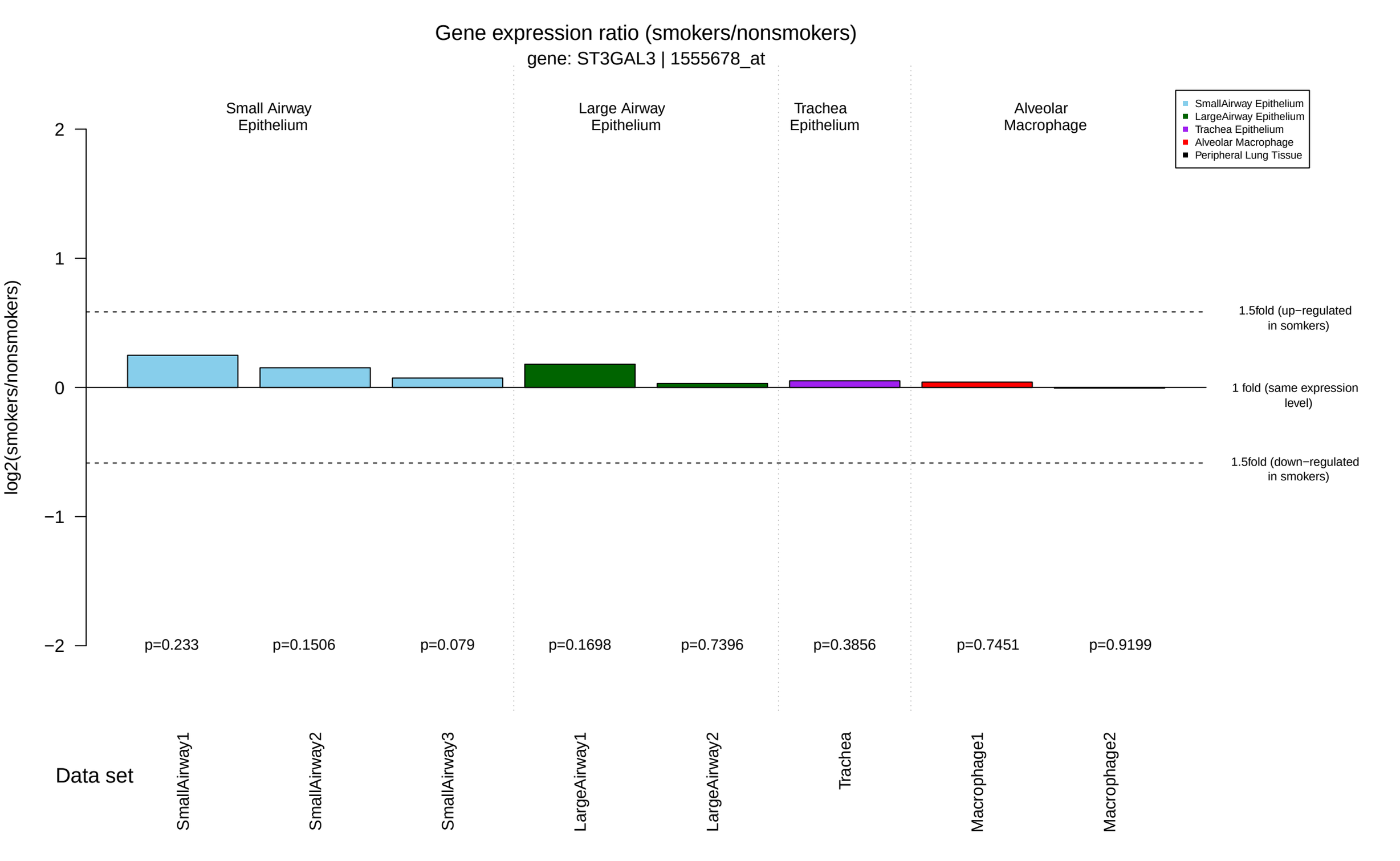
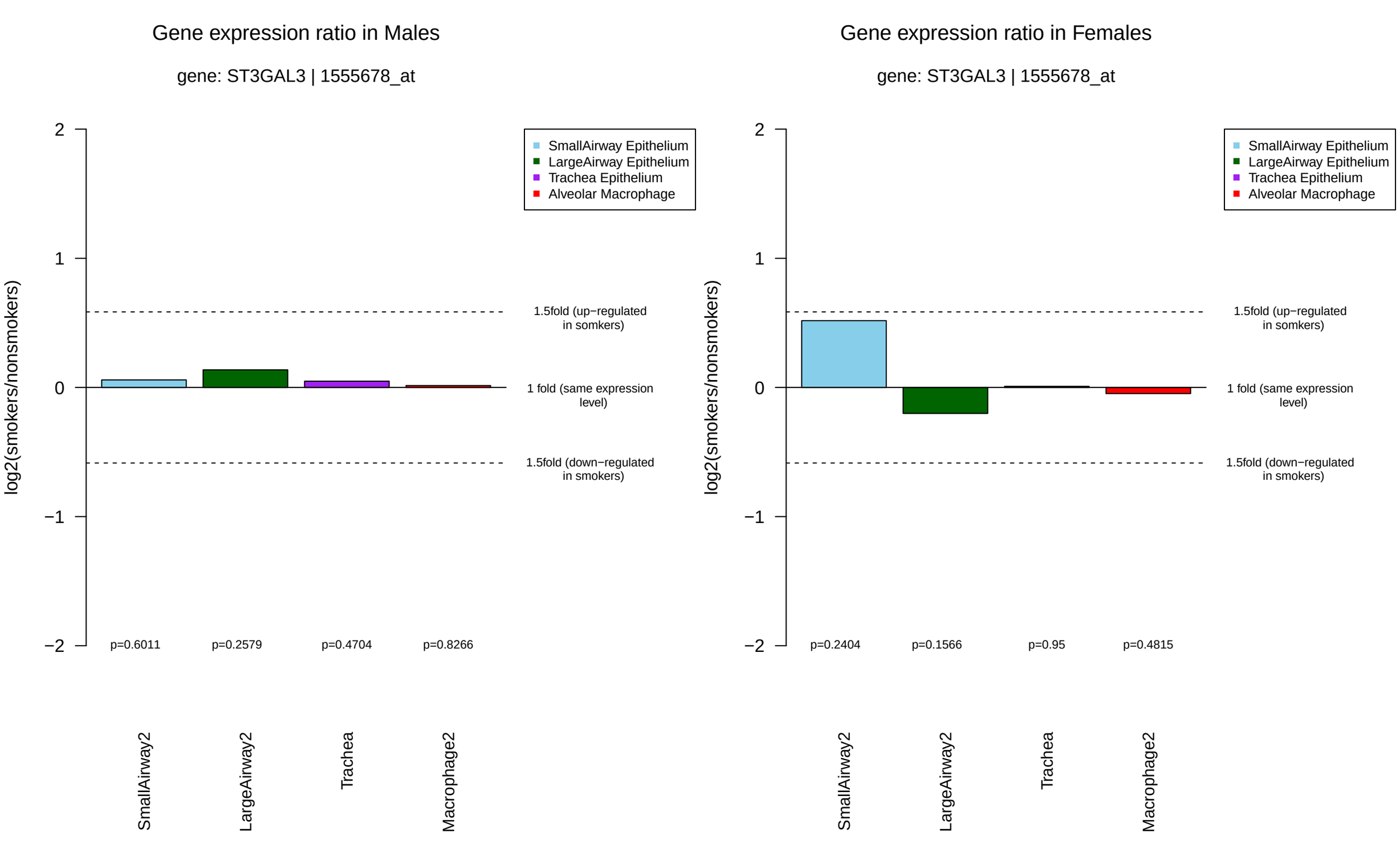
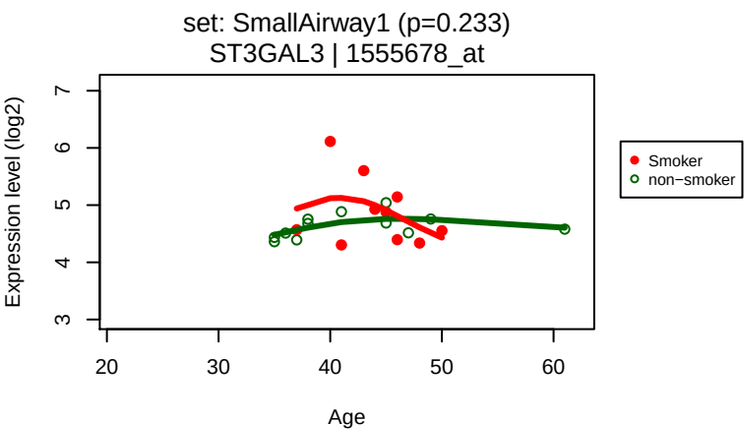
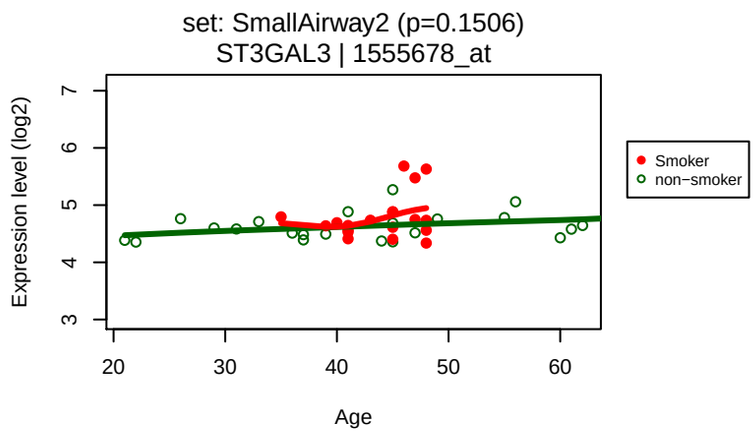
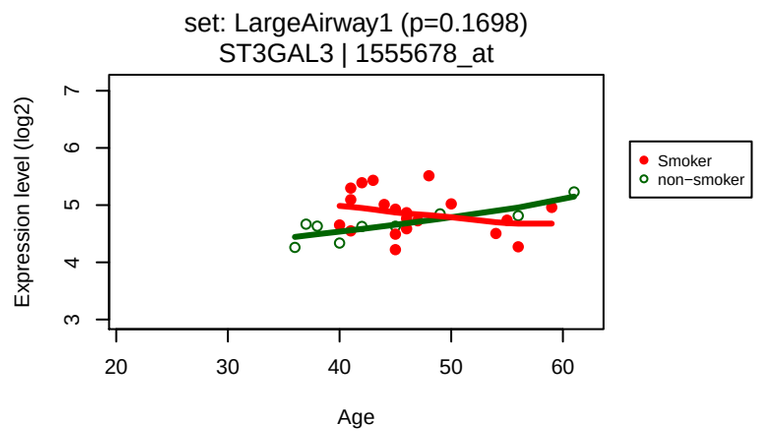

Gene expression of smokers and nonsmokers
gene: ST3GAL3 | 1555678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.233
p=0.1506
p=0.079
p=0.1698
p=0.7396
p=0.3856
p=0.7451
p=0.9199
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

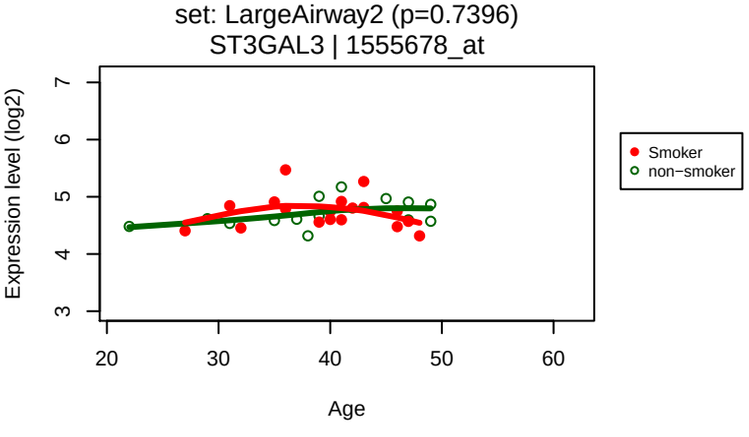

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL3 | 1555678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.233
p=0.1506
p=0.079
p=0.1698
p=0.7396
p=0.3856
p=0.7451
p=0.9199
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

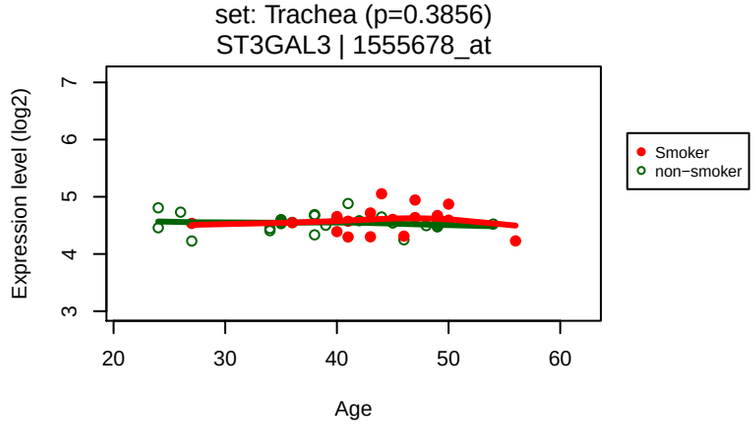

Gene expression ratio in Males
gene: ST3GAL3 | 1555678_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6011
p=0.2579
p=0.4704
p=0.8266
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL3 | 1555678_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2404
p=0.1566
p=0.95
p=0.4815
SmallAirway2
LargeAirway2
Trachea
Macrophage2

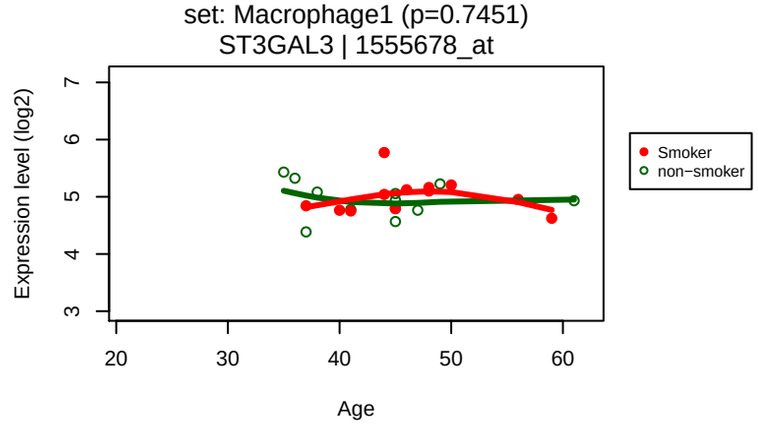

set: SmallAirway1 (p=0.233)
ST3GAL3 | 1555678_at
Expression level (log2)
Age
Smoker
non−smoker

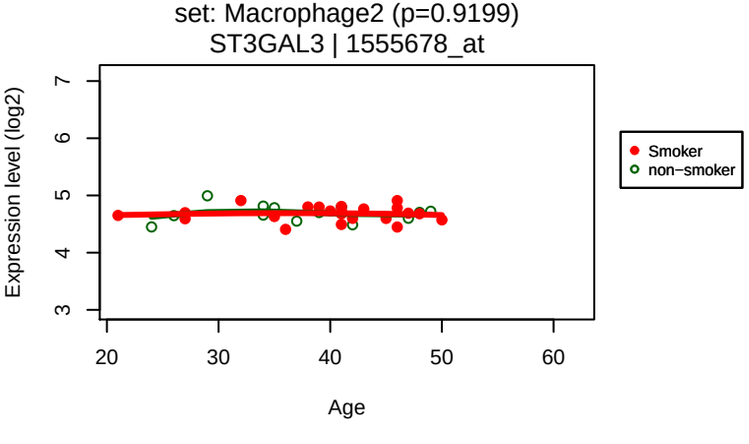

set: SmallAirway2 (p=0.1506)
ST3GAL3 | 1555678_at
Expression level (log2)
Age
Smoker
non−smoker

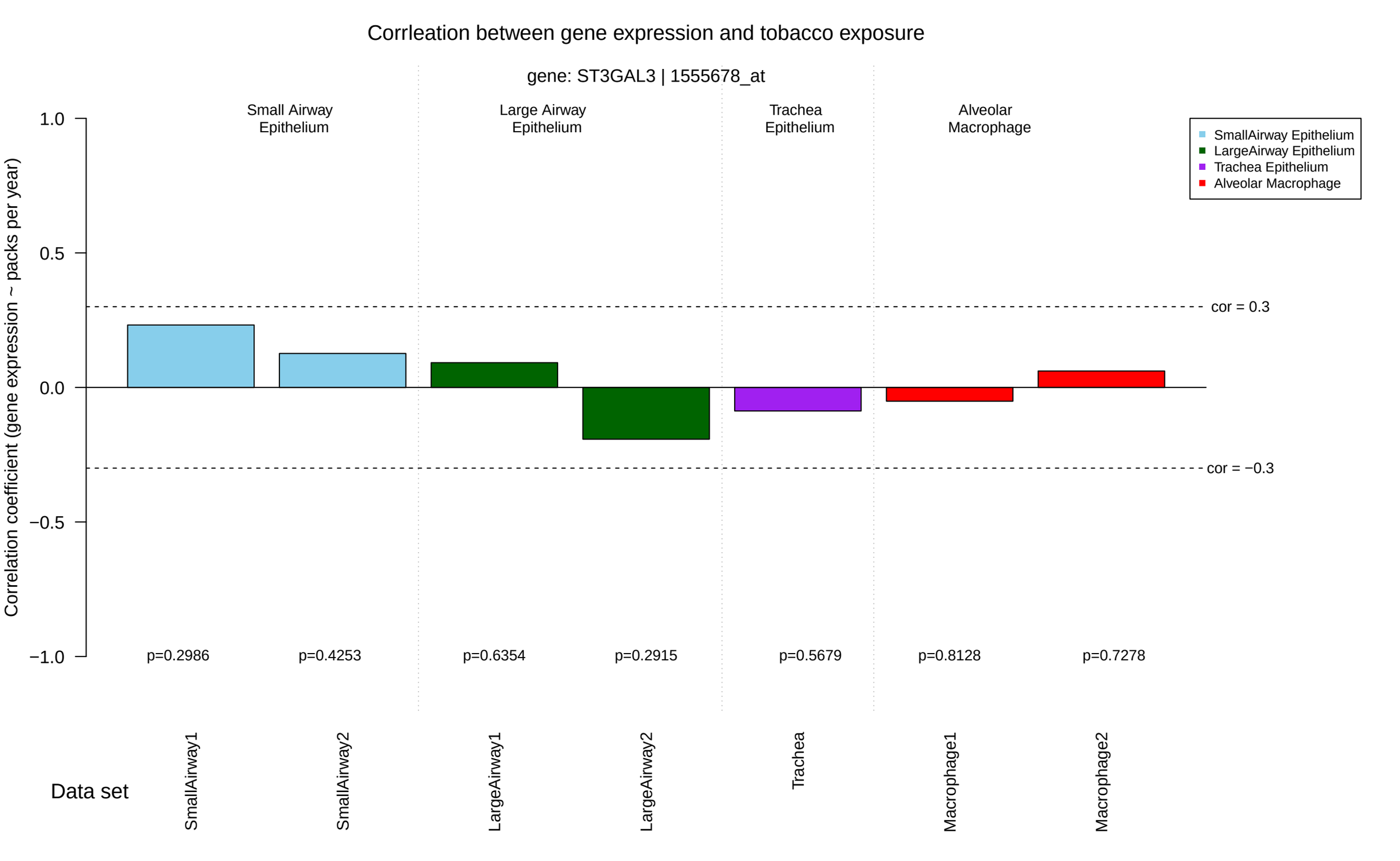

set: LargeAirway1 (p=0.1698)
ST3GAL3 | 1555678_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7396)
ST3GAL3 | 1555678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3856)
ST3GAL3 | 1555678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7451)
ST3GAL3 | 1555678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9199)
ST3GAL3 | 1555678_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL3 | 1555678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2986
p=0.4253
p=0.6354
p=0.2915
p=0.5679
p=0.8128
p=0.7278
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2